

DECENTRALIZED FEEDBACK STRUCTURES OF A VAPOR COMPRESSION CYCLE SYSTEM

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ABSTRACT

In vapor compression cycle (VCC) systems, it is desirable to effectively control the thermodynamic cycle. By controlling the thermodynamic states of the refrigerant with an inner-loop, supervisory algorithms can manage critical objectives such as maintaining superheat and maximizing the coefficient of performance, etc. In the HVAC industry, it is generally preferred to tune multiple single-input-single-output (SISO) control inner-loops rather than a single multiple-input-multiple-output (MIMO) control inner-loop. This paper presents a process by which a simplified feedback control structure amenable to a decoupled SISO control loop design may be identified. In particular, the many possible candidate input-output pairs for decentralized control are sorted via a decoupling metric, the relative gain array number. From a reduced set of promising candidate input-output pairs, engineering insight is applied to arrive at the final pairings successfully verified on a refrigeration test stand.

of performance (COP), a measure of system efficiency, is defined as the ratio between these two changes in enthalpy.

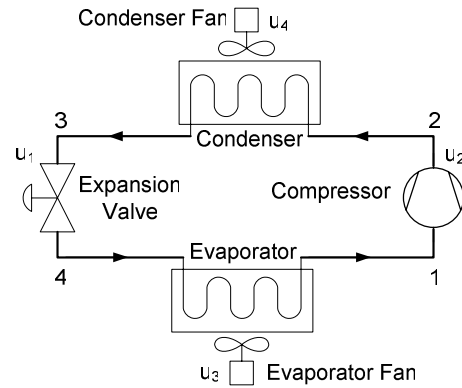


Figure 1: Schematic of ideal subcritical VCC system.

1. INTRODUCTION

The primary goal of any air conditioning or refrigeration application is to move energy from one location to another. An idealized vapor compression cycle (VCC) system, as shown in Figure 1 and Figure 2, is a thermodynamic system driven by the phase characteristics of the refrigerant that is flowing through it. An ideal VCC system assumes isentropic compression, isenthalpic expansion, and an isobaric process across both the condenser and evaporator coils. The basic control objectives of a VCC system can be conceptualized visually via Figure 2. For example, the difference between h_1 and h_4 represents the increase in enthalpy across the evaporator, i.e. the amount of energy (Q) removed from the cooled space. This is a measure of evaporator capacity. The difference between h_2 and h_1 represents the increase in enthalpy across the compressor, i.e. the amount of work (W) done by the compressor to increase the pressure of the refrigerant vapor. The system coefficient

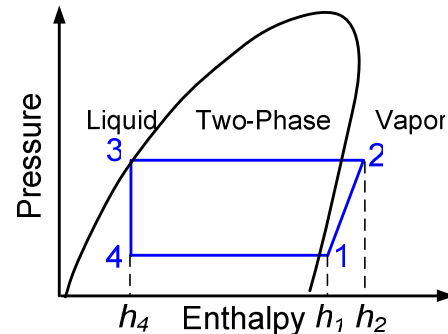


Figure 2: P-h diagram for ideal subcritical VCC.

$$COP = \frac{|Q|}{W} \approx \frac{h_4 - h_1}{h_2 - h_1} \quad (1)$$

Rather than explicitly controlling capacity or efficiency, however, this paper aims to control individual thermodynamic states of the system. Given the assumptions made for an ideal VCC system and the constitutive relationships between pressure, temperature, and enthalpy (i.e. for a given point on the cycle, only two thermodynamic states are required to derive the remaining states at that point), only four thermodynamic states are needed to uniquely define the four critical points of the idealized cycle shown on Figure 2. With an appropriate control architecture, these four points of the VCC could be placed by a higher level planning algorithm [1] so as to achieve an optimal balance between desired capacity and efficiency (see Figure 3). This is a shift from current practice where the actuators are used to control specific control objectives. For example, in [1] the compressor is used to control capacity and the expansion valve is used to control evaporator superheat.

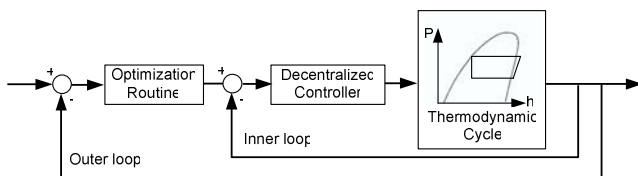


Figure 3: Block diagram visualization of decentralized controller within a larger optimization control loop.

It has been shown that multivariable control techniques [2][3][4] can be used to handle input-output couplings while achieving desired performance objectives. In [2] the system superheat and evaporator saturation temperature were used as the measured outputs in a Linear Quadratic Gaussian (LQG) control approach. However, for industrial practitioners and service engineers in the HVAC industry, it is generally preferred to tune multiple single-input-single-output (SISO) control loops rather than a single multiple-input-multiple-output (MIMO) control loop. Decentralized control structures consisting of SISO control loops will be greatly simplified when the interaction among the individual SISO control loops is sufficiently minimized.

Numerous control schemes have been developed with superheat and evaporation temperature (or pressure) as the feedback signals [2][3][4][5]. These references frequently noted the difficulty of controlling the two outputs with individual SISO control loops due to the physical coupling between superheat and evaporation temperature. Most recently, Keir [6] suggested that a novel choice of output control variables can decouple the dynamics such that a decentralized control approach can be used to meet desired performance objectives. As motivated by [6], this paper focuses on a structured method for identifying control input and output pairs which effectively decouple system dynamics. The output thermodynamic states considered for the control I/O pair analysis are derived from the four labeled points of the idealized cycle in Figure 2.

The rest of the paper is as follows. Section 2 introduces the choice of candidate input control variables and output thermodynamic states and describes the generation of identified models used for the ensuing analysis. The next section describes the use of a decoupling metric to sort through candidate input-output pairings. Section 4 discusses the results of the analysis and narrows the field of candidate I/O pairings to a final set. Finally, Section 5

provides a comparison between similar feedback controller algorithms applied to the best pairings of input parameters and control variables and to traditional I/O pairings.

2. SYSTEM MODELING AND IDENTIFICATION

While first principles models for VCC systems are available, Rasmussen [7] verified that system identification (ID) techniques can be used to construct accurate, low-order, local linear models of VCC systems. In this paper, the basic dynamic response of a VCC is identified using a time domain system ID procedure. Four controllable inputs for a variable-speed VCC are considered: the compressor speed, expansion valve opening, evaporator fan speed, and condenser fan speed (see u_1 through u_4 in Figure 1). This is often not the case, due to the fact that in many cases, e.g. automotive systems, the condenser air flow is a function of vehicle speed and acts more as a disturbance to the feedback loop. Nevertheless, they are considered here to allow for a thorough analysis of I/O pairs.

There are six thermodynamic states that are considered as output measurements: two pressures and four temperatures. Recall from Figure 2 that, for an idealized cycle, there are two system pressures: $P_2=P_3$ and $P_1=P_4$. These correspond to the pressure inside the condenser and the pressure inside the evaporator, respectively. There are four system refrigerant temperatures: T_1 , T_2 , T_3 , and T_4 . Again assuming an idealized cycle with saturated refrigerant leaving the condenser, these represent evaporator outlet temperature, condenser inlet temperature, condenser saturation temperature, and evaporator saturation temperature, respectively.

The output responses to step changes in each of the four control inputs, as well as to random Gaussian combinations of all four inputs (see Figure 5) around a set of nominal operating conditions, were collected on an A/C experimental test stand (see Figure 4). For a more detailed description of the experimental system, see [6].



Figure 4: Experimental test stand at the University of Illinois at Urbana-Champaign.

A standard prediction error/maximum likelihood system identification algorithm in the System Identification Toolbox [8] was used to identify the frequency response between each input and output pair. Because system identification is sensitive to scaling, two models were identified, wherein the output parameters within each model shared the same units. That is, the first model was identified

with all four excited inputs and the two pressure measurements as outputs ($P_2=P_3$ and $P_1=P_4$) and a second model was identified with the same inputs and the four temperature measurements as outputs (T_1, T_2, T_3 , and T_4). As will be described in the next section, only the frequency response between each I/O pair is needed for the decoupling analysis, thereby allowing the flexibility to identify two separate models rather than a single 4-input 6-output model.

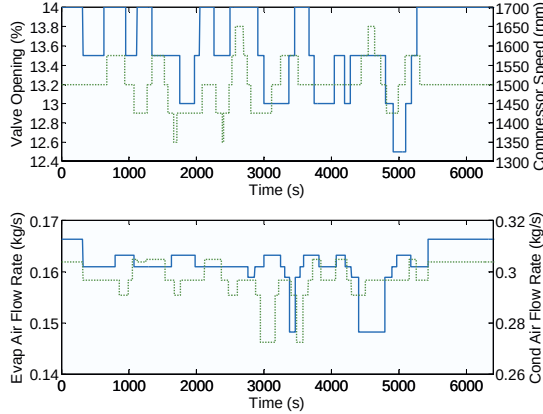


Figure 5: Random Gaussian input signals. Note: dashed signal corresponds to right-hand y-axis.

Figures 6 and 7 show the system identification results for each of the two identified models, respectively. The complete state space representation of each identified model is included in Appendix A. Note that using $P_{2-1} \triangleq P_2 - P_1$ instead of P_2 provided a better fit with respect to the system identification.

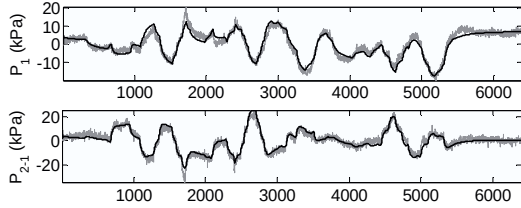


Figure 6: Detrended system ID results for P_1 : fit = 69.9%; P_{2-1} : fit = 71.25%.

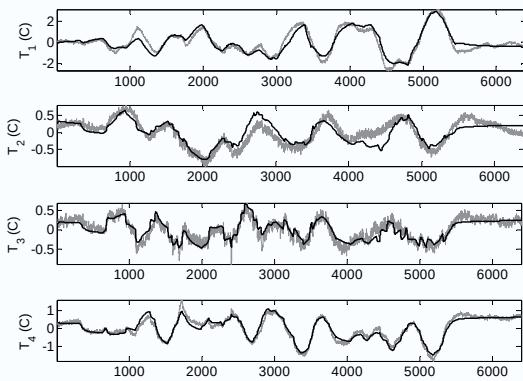


Figure 7: Detrended system ID results for T_1 : fit = 65.22%; T_2 : fit = 43.99%; T_3 : fit = 54.62%; T_4 : fit = 72.81%.

For each identified output, the open-loop response is compared against the response as predicted by the identified model. The fit percentages for each output characterizing the goodness of the identification are included in the captions of Figure 6 and Figure 7.

3. I/O PAIRING ANALYSIS

The following section applies a model-based decoupling metric to the above identified models in order to reduce the possible pairs to the most promising I/O pairs.

3.1 Output Combinations

For an idealized VCC, the output measurements $T_1, T_2, T_3, T_4, P_2=P_3$, and $P_1=P_4$ may be used to fully characterize the thermodynamic states of the cycle. However, other physically meaningful aspects of the cycle, such as evaporator superheat,

$$T_{1-4} = T_1 - T_4, \quad (2)$$

may present opportunities for further decoupling. In fact, there is an infinite number of affine combinations, represented by

$$\tilde{y} = \sum_{i=\{1,2\}} \alpha_i P_i + \sum_{i=\{1,2,3,4\}} \beta_i T_i \quad (3)$$

for some $\alpha_i, \beta_i \in \mathbb{R}$, that could be considered. Even though such affine combinations could be used to thoroughly decouple the individual identified loops, they are not necessarily physically meaningful in terms of the VCC. The inverse and singular value decomposition (SVD) of the DC-gain could be used to determine affine combinations that decouple the individual loops, but such combinations are sensitive to the choice of input and output scaling [9] and disregard the physical significance and units of the resulting combinations. For example, the units of $\tilde{y} = \alpha_1 P_1 + \beta_1 T_1$ are ambiguous, and the simple addition of a pressure and temperature does not carry any particular physical significance. Instead, for this investigation, each output is defined as a binary combination of either the temperatures or pressures, representing differences and averages, such as evaporator superheat or pressure differential. Each output combination $\tilde{y}$ takes the form of (3), where $(\beta_i = 0, \alpha_i = \{-1, 0, 1\})$ or $(\beta_i = \{-1, 0, 1\}, \alpha_i = 0)$.

There is a very large number of potential input-output pairing combinations that may be effective in decoupling the individual loop. The possible binary combinations (3) for either two pressures and four temperatures are $\sum_{i=1}^2 3^{(i-1)} = 4$ and $\sum_{i=1}^4 3^{(i-1)} = 40$, respectively. Consequently, there are $44!/(44-4)! = 3,258,024$ possible I/O pairs when considering 4 inputs to the system.

3.2 Sorting via a Decoupling Metric

Given the large number of possible input-output pairings, it is essential that a decoupling metric be used to filter them. One metric that can be used to quantify the reduction in coupling is the relative gain array (RGA) technique originally developed by Bristol [10]. The technique is independent of input and output scaling, thereby avoiding the question of appropriate scaling of inputs and outputs

with different magnitudes. To apply the RGA analysis, a dynamic model of the system must be identified using time domain data as was described in Section 2. The RGA is a steady-state measure of closed loop interactions for decentralized (multiple SISO loop) control. For a non-singular square matrix P , the relative gain array is defined by Equation (4), where $\times$ denotes the Schur product.

$$RGA(P) = \Lambda(P) = P \times (P^{-1})^T \quad (4)$$

The RGA is a good indicator of sensitivity to uncertainty in the input channels, diagonal dominance for state-space systems, and stability levels of decentralized controller designs [9].

Uncertainty in the input channels is indicated by plants with large RGA elements around the crossover frequency, making these plants fundamentally difficult to control. A measure of the diagonal dominance of a plant, G , is obtained by calculating the RGA-number, given in Equation (5).

$$RGA\text{-}number(G(j\omega)) = \|\Lambda(G(j\omega)) - I\|_{sum}, \quad (5)$$

where $\|M\|_{sum}$ for some square matrix $M \in \mathbb{R}^{n \times n}$ is defined as

$$\sum_{i=1}^n \sum_{j=1}^n |m_{ij}|.$$

Large RGA numbers are a clear indicator that the closed loop performance will be poor when decentralized control schemes are applied [9]. When all four inputs are considered in the RGA analysis, the best achievable RGA-number is 1.4374 (see Figure 8).

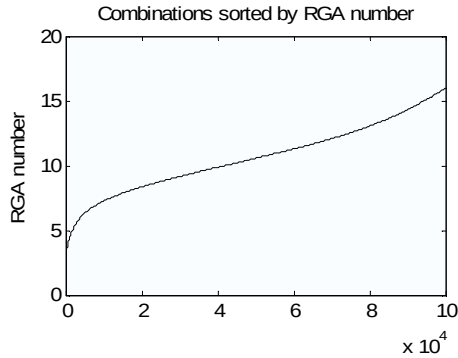


Figure 8: Variation of RGA number for the best 100,000 (of ~3,258,024) candidates for 4 inputs.

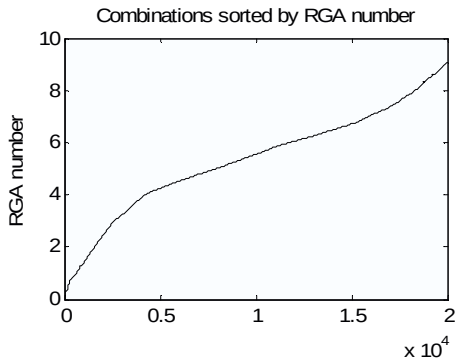


Figure 9: Variation of RGA number for the best 20,000 (of ~76464) candidates for 3 inputs.

When the condenser air flow rate u_4 is removed from the group of inputs considered, the best achievable RGA-number is 0.076 (see Figure 9), a 20-fold reduction compared to the minimum RGA-number for the four input case. Based on this result, we consider only inputs u_1 through u_3 in the analysis presented in Section 4.

4. COMPARISON OF COUPLING FOR DIFFERENT I/O PAIRINGS

Applying the RGA number as a decoupling metric, the control designer is presented with a reduced field of the best potential I/O pair candidates. Subsequently, physical intuition and an understanding of vapor compression cycles is applied to further narrow the field for experimental trials. The following tables characterize how frequently each output was associated with an input u_i for the top 500 I/O pairs, as characterized by the RGA-number, at three different frequencies. Note the tables only present the top recurring outputs coupled to each input; therefore the rows do not necessarily total 100%. Moreover, recall that u_1 , u_2 , and u_3 represent EEV opening, compressor speed, and evaporator airflow rate, respectively.

TABLE 1: INCIDENCE OF CANDIDATE PAIRINGS FOR INPUT U_1 FOR TOP 500 PAIRINGS

freq.	$P_2=P_3$	T_3	P_{3+4}	T_{3+4}
$\omega=0$ rad/s	64%	26%	6%	0%
$\omega=0.01$ rad/s	30%	7%	0%	6%
$\omega=1$ rad/s	0%	10%	2%	7%

TABLE 2: INCIDENCE OF CANDIDATE PAIRINGS FOR INPUT U_2 FOR TOP 500 PAIRINGS

freq.	$P_1=P_4$	P_{2-4}	T_{2-1}	T_{2-4}
$\omega=0$ rad/s	26%	6%	9%	11%
$\omega=0.01$ rad/s	6%	52%	9%	6%
$\omega=1$ rad/s	2%	6%	3%	0%

TABLE 3: INCIDENCE OF CANDIDATES OUTPUTS FOR U_3 FOR TOP 500 PAIRINGS

freq.	$T_{1+2+3+4}$	T_{1+2+4}	T_{1+3+4}	$T_{1+2-3+4}$	T_{1+4}
$\omega=0$ rad/s	8%	6%	6%	5%	6%
$\omega=0.01$ rad/s	8%	5%	6%	3%	4%
$\omega=1$ rad/s	3%	10%	9%	3%	9%

The statistics presented in Tables 1-3, and an understanding of the vapor compression cycle, are applied to further develop engineering insight for selecting candidate I/O pairings. For example, Table 1 suggests the EEV, input u_1 , most strongly drives the condenser pressure, $P_2(=P_3)$, particularly at low frequencies. Figure 10 provides a qualitative metric for assessing how strongly u_1 drives each of these candidate output states. Recall that the experimental system was perturbed with a series of step changes in each input; here we show a particular output state against only one input signal at a time to more clearly assess the coupling between that I/O pair.

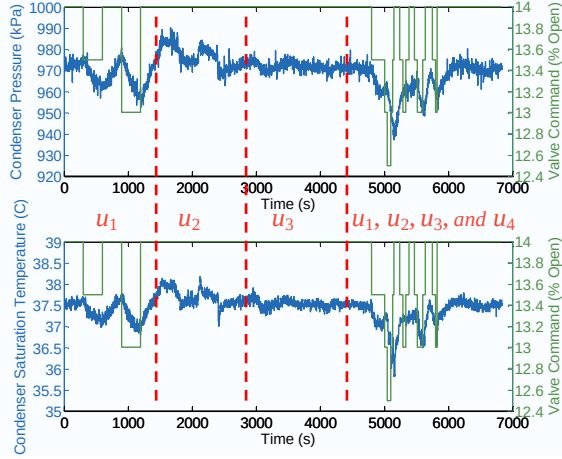


Figure 10: Open loop response of I/O coupling for input u_1 .

Note that the response of both $P_2=P_3$ and T_3 are relatively slow, which is consistent with the RGA analysis which showed a stronger coupling at lower frequencies. This pairing also agrees with the intuition that the EEV controls the refrigerant mass flow rate leaving the condenser. Oftentimes, the EEV is used to regulate the evaporator superheat T_{1-4} , because the electronic expansion valve is viewed as a programmable version of another device, a thermal expansion valve, used to regulate evaporator superheat. Consequently, regulation of the condenser pressure via the EEV runs somewhat counter to the typical VCC control design practice. Gaining potentially unconventional insights is one of the benefits of sorting through such a large set of I/O pairings.

Table 2 suggests that the compressor speed, input u_2 , is tightly coupled with both the evaporator pressure, $P_1 (=P_4)$, and the difference between the condenser and evaporator pressures, P_{2-1} . In particular, the RGA analysis suggests that u_2 is more strongly coupled with P_1 at low frequencies, but that at higher frequencies, it drives P_{2-1} . The open loop response for these I/O pairings is provided in Figure 11.

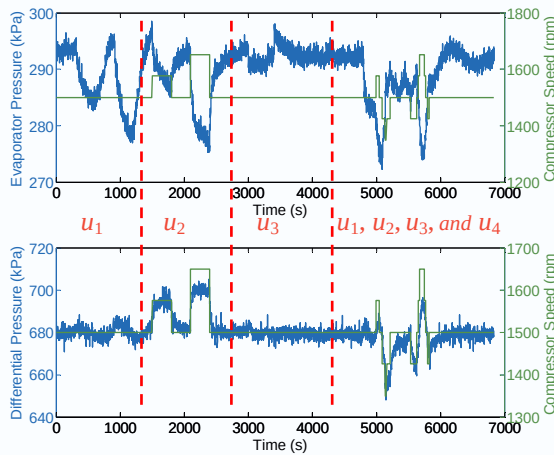


Figure 11: Open loop response of I/O coupling for input u_2

Again there is a strong correlation between the frequency-based RGA analysis and the open loop responses presented in Figure 11.

The response of $P_1(=P_4)$ is slower than that of P_{2-1} , confirming that there is tighter coupling at lower frequencies between u_2 and $P_1(=P_4)$, but that at high frequencies, the coupling is stronger between u_2 and P_{2-1} . These two particular I/O parameter pairs were originally explored in [6], wherein the reduction in coupling exhibited by using P_{2-1} rather than P_1 was shown to significantly improve the performance of the decentralized two-input-two-output closed loop system. However, only two inputs were considered in [6], and the best choice of I/O pairs might be different when considering three inputs.

Table 3 suggests that the evaporator airflow rate, u_3 , drives some combination of the four temperatures. Both the evaporator outlet and evaporator saturation temperatures consistently appear in the most prominent combinations in Table 3, suggesting that an average of those two alone might be a good candidate. Again these candidate I/O pairings can be evaluated qualitatively using experimental data of the open loop response of each to perturbations in u_3 .

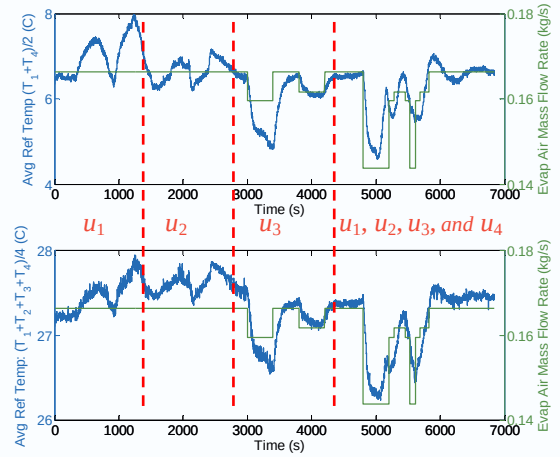


Figure 12: Time plot of I/O coupling for input u_3

From Figure 12 both an average of T_1 and T_4 , as well as an average of all four temperatures, appear to respond similarly to perturbations in u_3 . From a physical perspective, this is particularly interesting because controlling the average evaporator refrigerant temperature would in effect allow the control designer to regulate evaporator superheat which is critical to ensuring that the refrigerant entering the compressor is in the vapor form.

With the field of possible I/O pairings narrowed down to two candidate outputs for each input (see Figure 13), one final metric is needed to better understand if one I/O pair provides an advantage over another. Note that $T_{1+2+3+4}$ has been replaced with T_1 . While there is some loss in decoupling, considering T_1 is more practical, from a systems perspective, as a parameter that is meaningful to the vapor compression cycle.

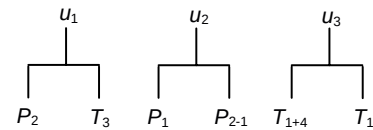


Figure 13: Final I/O candidate pairs.

Because the RGA analysis is frequency-based, it is important to more closely examine how the coupling between each I/O pair varies with respect to frequency. Tables 1-3 suggested that certain output states were more strongly coupled at either low frequencies or high frequencies, but not necessarily both. Given two possible output states for each control input u , there are eight possible I/O pairing sets that can be used to control the system. Table 4 lists these eight sets. Figures 14 and 15 characterize the RGA-number for each I/O set as function of frequency.

TABLE 4: CANDIDATE PAIRINGS BETWEEN INPUT AND OUTPUTS

Inputs/ Outputs	Pair y_1	Pair y_2	Pair y_3	Pair y_4	Pair y_5	Pair y_6	Pair y_7	Pair y_8
u_1	P_3	P_3	P_3	P_3	T_3	T_3	T_3	T_3
u_2	P_{3-4}	P_{3-4}	P_4	P_4	P_{3-4}	P_{3-4}	P_4	P_4
u_3	T_{1+4}	T_1	T_{1+4}	T_1	T_{1+4}	T_1	T_{1+4}	T_1

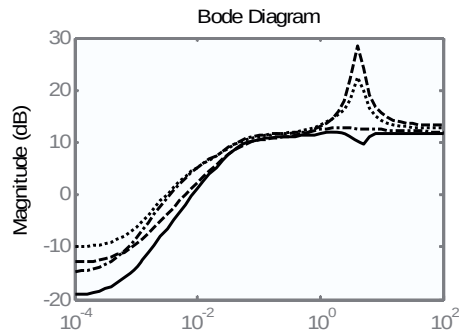


Figure 14: RGA number frequency plot; $y_1 = '-', y_2 = '--', y_3 = '-.', y_4 = '.'$

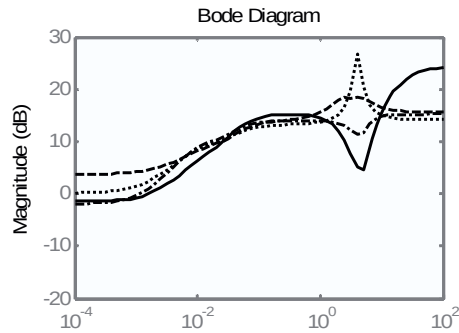


Figure 15: RGA number frequency plot; $y_5 = '--', y_6 = '--', y_7 = '-.', y_8 = '.'$

Sets $y_1 (= [P_3, P_{3-4}, T_{1+4}])$ and $y_3 (= [P_3, P_4, T_{1+4}])$ show the lowest RGA number at low frequencies and no spikes in RGA-number at higher frequencies, making both sets good candidates. Choosing y_1 as the optimal I/O set, the idea of controlling individual points of the cycle can be revisited. As an illustration, see Figure 16.

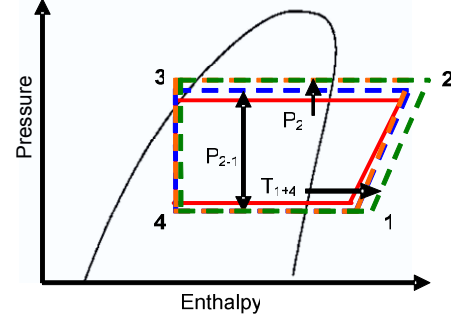


Figure 16: Idealized system schematic.

While parameters such as evaporator superheat and evaporator pressure might still be preferable in terms of describing desired performance, Figure 16 illustrates how the chosen I/O pairs still allow for these parameters to be controlled indirectly. For example, although a decentralized control approach would be used to control condenser pressure and differential pressure independently, a correct choice of reference signals for each would result in the indirect control of evaporator pressure. A similar argument can be made for evaporator superheat. This reinforces the importance of selecting I/O pairs which sufficiently decouple the system and using an outer control loop to coordinate those thermodynamic states (which can be controlled effectively with individual SISO loops) to achieve desired performance objectives.

5. COMPARISON OF DECENTRALIZED CONTROLLER DESIGNS

To further demonstrate the system performance with the candidate I/O pairings, the set $y_1 (= [P_3, P_{3-4}, T_{1+4}])$ is used to present a decentralized proportional-integral-derivative (PID) controller constructed with three individual SISO control loops.

The decentralized feedback configuration used for the controller design of the experimental system is shown in Figure 17. The controllers $K_1(s)$, $K_2(s)$, and $K_3(s)$ are independent PID controllers, since some types of PID algorithms are prevalent in the majority of industrial controllers for these kinds of systems. The output variables are the condenser pressure, P_3 , the difference between condenser and evaporator pressure, P_{3-4} , and the average of evaporator saturation and evaporator outlet temperatures, T_{1+4} .

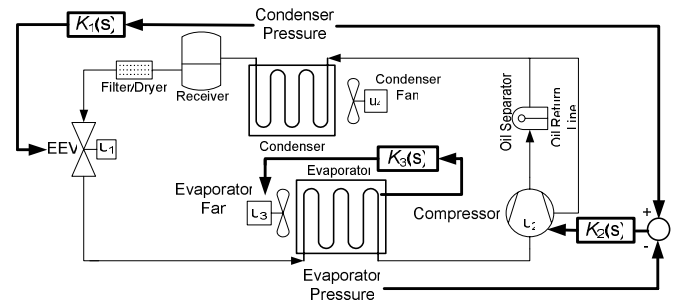


Figure 17: A control approach for vapor compression systems

PID controllers were tuned online to evaluate the closed loop system's reference tracking characteristics. This highlights the

industrial attractiveness of using PID algorithms, since the controllers can be tuned online during experiments and do not require extensive first principle models. A sequence of step input reference signals was applied to the condenser pressure, P_3 , and pressure differential, P_{3-4} , respectively, while the average evaporation temperature T_{1+4} was regulated at a constant set point. The three controller gains are given in Table 5.

TABLE 5: TUNED GAINS FOR THREE CONTROLLERS

Input/Controller	Proportional Gain	Integral Gain	Derivative Gain
EEV opening u_1/K_1	0.25	0.002	2
Compressor speed u_2/K_2	12	0.4	3
Evap. airflow rate u_3/K_3	3	0.05	20

Figure 18 shows the time domain response of the output of each control loop. The condenser pressure controller, $K_1(s)$, and pressure differential controller, $K_2(s)$, have a settling time of about 60 seconds, while the oscillation around the pressure set points is within ± 3 kPa. The average temperature feedback controller, $K_3(s)$, regulates the average temperature within $\pm 1^\circ\text{C}$ error during the tracking process. The results demonstrate the ability of three controllers to independently track three set points while selecting the set $y_1 (= [P_3, P_{3-4}, T_{1+4}])$ as the I/O pairing.

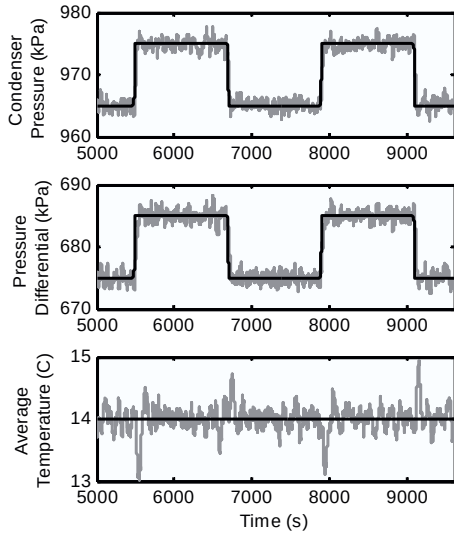


Figure 18: Time domain output response with the set $y_1 (= [P_3, P_{3-4}, T_{1+4}])$ feedback controller.

Next we provide a comparison of this candidate I/O pairing set with a baseline I/O pairings for this type of system. Most VCC systems utilize an electronic expansion valve (EEV) to regulate the superheat (T_{1-4}) of the system, while the compressor speed is used to control either evaporator temperature or evaporator pressure (T_4 and P_4 , respectively). All three inputs are not used because evaporator superheat is simply a function of evaporator temperature and pressure; that is, each output is uniquely determined by the other two. This suggests an additional advantage of the approach presented in this paper wherein the candidate output set chosen in Section 4

contains three independent parameters, allowing for an additional degree of freedom.

For the I/O pairing $y_{\text{baseline}} \triangleq [T_{1-4}, P_4]$ with EEV opening and compressor speed as inputs, two SISO PID controllers were tuned on the experimental test stand. The reference tracking results of these two SISO loops are shown in Figure 20. It is clear that the actuator input signals required by the controllers in Figure 19 result in less-oscillation compared to the EEV and compressor actuation shown in Figure 20. Similar results are also discussed in [6].

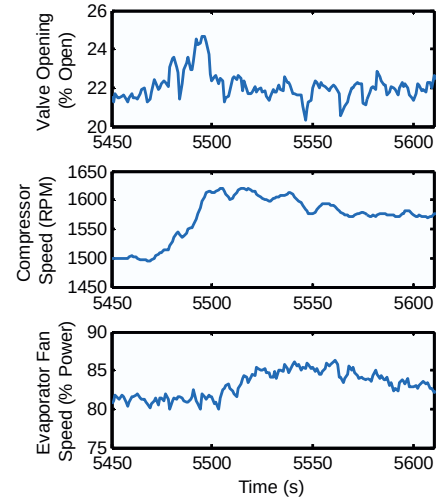


Figure 19: Actuator performance with $y_1 (= [P_3, P_{3-4}, T_{1+4}])$ feedback controller.

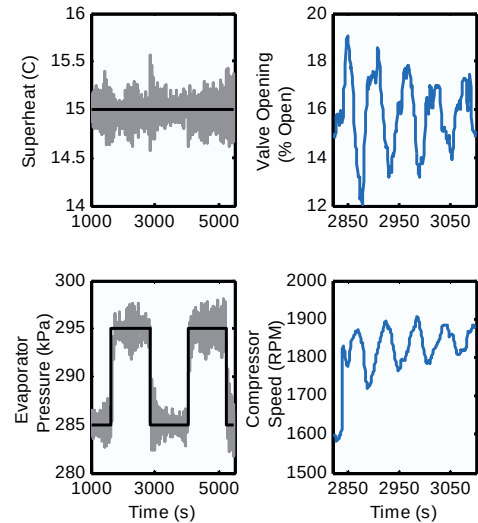


Figure 20: System performance with $y_{\text{baseline}} (= [T_{1-4}, P_4])$ feedback controller.

The results presented here suggests that a decentralized control approach using well chosen I/O pairings results in effective reference tracking of each output with less wear on system actuators. Moreover, this suggests that it is possible to design an outer control loop to choose the appropriate set points for each of the output states in order to meet desired performance objectives.

6. CONCLUSIONS

This paper demonstrates that an appropriate choice of feedback variables can improve the effectiveness of a decentralized controller on a VCC system in the HVAC industry. Rather than explicitly controlling capacity or efficiency, three individual SISO control loops are used to control specific thermodynamic states of the system while relying on a higher level planning algorithm to optimize the set points for each state such that desired performance and efficiency objectives can be met. An expansive set of I/O pairings was considered, and a decoupling metric and engineering insight were used to choose the best pairs.

It was found that the EEV is best paired with condenser pressure, the compressor speed with differential pressure, and the evaporator fan with average evaporator temperature. The primary advantage of considering such a large set of I/O pairings was that it led to potentially unconventional insights with regards to optimal pairings for decentralized control. It should be noted that the analysis performed here is for a linear system representation. VCC systems are actually very nonlinear systems. Any linear VCC system representation will vary greatly in its parameters about different operating points [12]. However, linearizing the system about any operating point and performing a local analysis will result in a similar type of decoupling benefit as that presented in this paper.

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APPENDIX A. IDENTIFIED STATE SPACE MODELS

The identified state space $[A, B, C, D]$ system model for two output pressures is given in Equation (6). Note that differential pressure, rather than P_2 , was used for the identification.

$$\begin{aligned} A &= \begin{bmatrix} -0.06794 & 0.04704 & 0.2834 & -0.0481 & 0.2248 & 1.042 \\ -0.05117 & -0.2241 & -0.03257 & -0.2986 & 1.154 & 1.587 \\ 0.2587 & -0.3095 & -1.295 & 0.05653 & -0.4164 & -3.854 \\ -0.245 & -0.22 & 0.7531 & -0.7526 & 2.883 & 6.105 \\ 0.6989 & -3.387 & -5.924 & -2.978 & -0.9685 & -5.691e-6 \\ -0.5499 & 0.8788 & 2.963 & 0.1192 & 0 & -1.327 \end{bmatrix} \\ B &= \begin{bmatrix} -0.001744 & 4.635e-6 & 0.1389 & 0.02843 \\ -0.002344 & -4.676e-5 & 0.2562 & 0.06265 \\ 0.007939 & -3.87e-5 & -0.4873 & -0.1302 \\ -0.01288 & -9.731e-6 & 0.9648 & 0.2149 \\ 0.002021 & -0.0004335 & 0.7811 & 0.06284 \\ -0.01574 & 0.0001202 & 0.8895 & 0.2418 \end{bmatrix} \\ C &= \begin{bmatrix} 256.3 & 93.39 & 14.05 & -51.34 & 0 & 0 \\ 95.29 & -237 & 49.71 & 68.76 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \quad (6)$$

$$u = \begin{bmatrix} a_v & \omega_c & \dot{m}_{e,air} & \dot{m}_{c,air} \end{bmatrix}^T \quad y = \begin{bmatrix} P_1 & P_{2-1} \end{bmatrix}^T$$

The identified state space $[A, B, C, D]$ system model for the four output temperatures is given in Equation (7).

$$\begin{aligned} A &= \begin{bmatrix} -0.005231 & 0.001287 & 0.001944 & 0.001778 \\ -0.004315 & -0.009934 & -0.0007708 & -0.01032 \\ -0.002326 & -0.01244 & -0.02806 & 0.01421 \\ -0.00884 & -0.03749 & -0.008136 & -0.09927 \end{bmatrix} \\ B &= \begin{bmatrix} -0.0002647 & -1.453e-6 & 0.02311 & 0.004852 \\ -0.0003534 & 1.08e-5 & 0.02181 & -0.001921 \\ -0.0006485 & 4.105e-6 & -0.1572 & 0.01233 \\ -0.001648 & 7.498e-5 & 0.1627 & -0.05476 \end{bmatrix} \\ C &= \begin{bmatrix} 140.8 & 9.181 & -13.99 & 2.757 \\ -13.7 & 19.41 & -6.834 & 0.3653 \\ -6.119 & -0.3302 & -2.125 & 3.259 \\ -10.9 & -5.542 & -4.846 & -0.8949 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ u &= \begin{bmatrix} a_v & \omega_c & \dot{m}_{e,air} & \dot{m}_{c,air} \end{bmatrix}^T \quad y = \begin{bmatrix} T_1 & T_2 & T_3 & T_4 \end{bmatrix}^T \end{aligned} \quad (7)$$